

Government Intervention in Markets: Price Controls, Taxes, Subsidies & Quotas. *Part I*

ECON 101: Introduction to
Microeconomics

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Government Interventions in Markets

- Price Controls
 - Price Ceilings
 - Price Floors
- Taxes
- Subsidies
- Production Quotas
- Markets for Illegal Goods



Price Ceilings

A **price ceiling** or **price cap** is a regulation that makes it illegal to charge a price higher than a specified level.

When a price ceiling is applied to a housing market it is called a **rent ceiling**.

If the rent ceiling is set *above* the equilibrium rent, it has no effect. The market works as if there were no ceiling.

But a rent ceiling set *below* the equilibrium rent creates

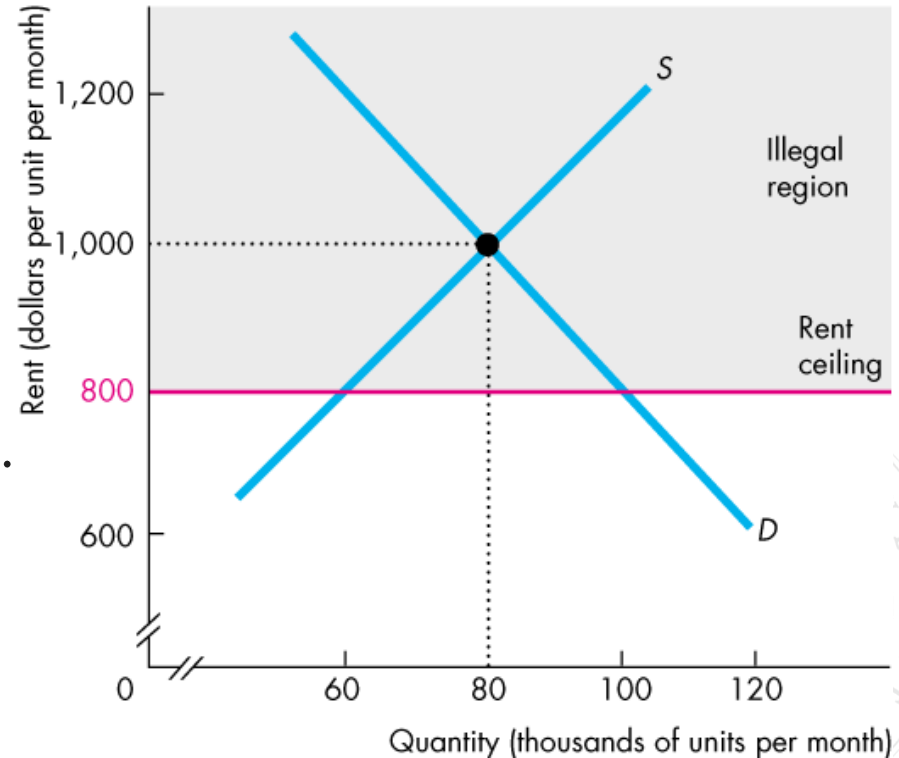
- A housing shortage
- Increased search activity
- A black market



A housing market with a Rent Ceiling

Housing Shortage

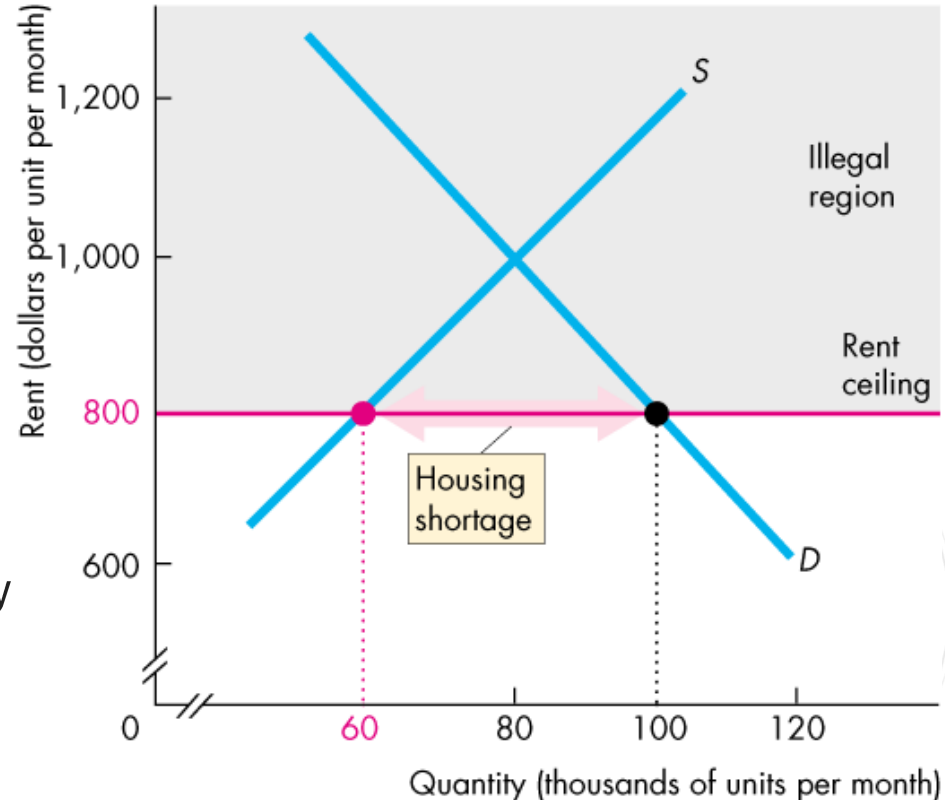
- The figure shows the effects of a rent ceiling that is set *below* the equilibrium rent.
- The equilibrium rent is \$1,000 a month.
- A rent ceiling is set at \$800 a month.
- So the equilibrium rent is in the illegal region.



A housing market with a Rent Ceiling

Housing Shortage

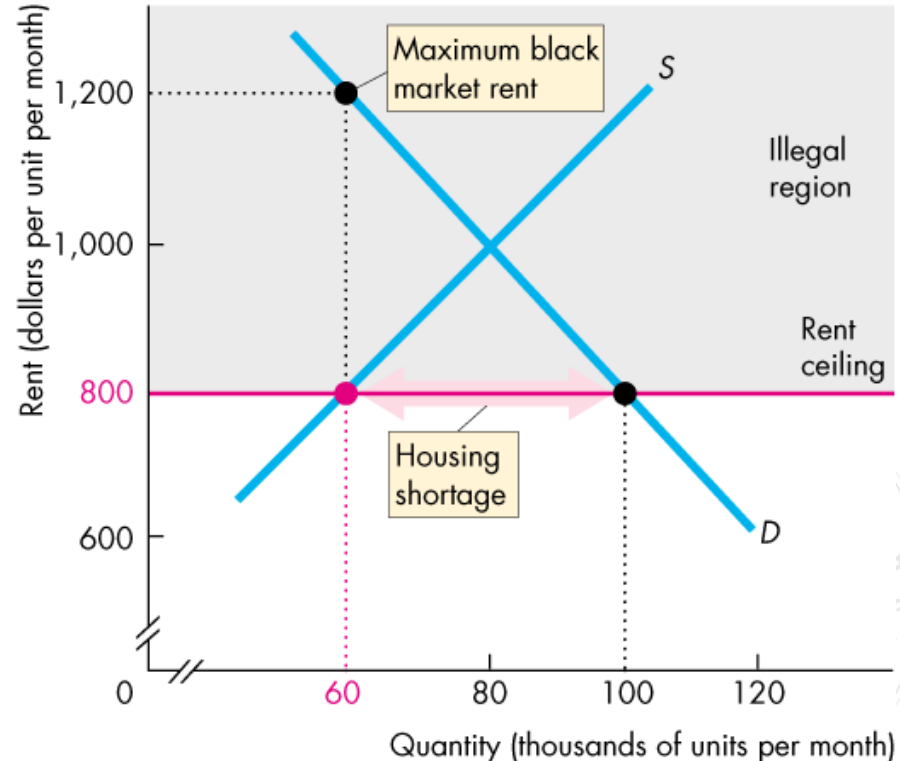
- The figure shows the effects of a rent ceiling that is set *below* the equilibrium rent.
- The equilibrium rent is \$1,000 a month.
- A rent ceiling is set at \$800 a month.
- So the equilibrium rent is in the illegal region.
- At the rent ceiling, the quantity of housing demanded exceeds the quantity supplied.
- There is a shortage of housing.



A housing market with a Rent Ceiling

Other mechanism operate to determine the price

- Because the legal price cannot eliminate the shortage, other mechanisms operate:
 - Increased search activity
 - A black market
- With the shortage someone is willing to pay up to \$1200 a month.



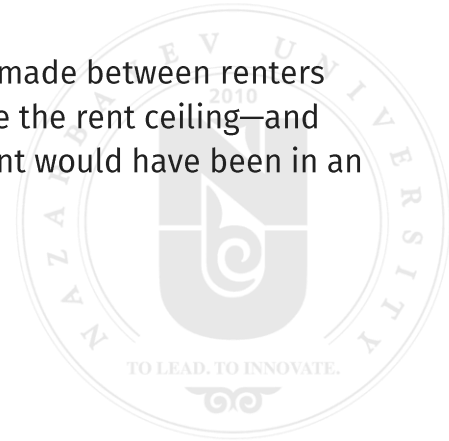
A Housing Market with a rent ceiling

Increased Search Activity

- The time spent looking for someone with whom to do business is called **search activity**.
- When a price is regulated and there is a shortage, search activity increases.
- Search activity is costly and the opportunity cost of housing equals its rent (regulated) plus the opportunity cost of the search activity (unregulated).
- Because the quantity of housing is less than the quantity in an unregulated market, the opportunity cost of housing exceeds the unregulated rent.

A black market

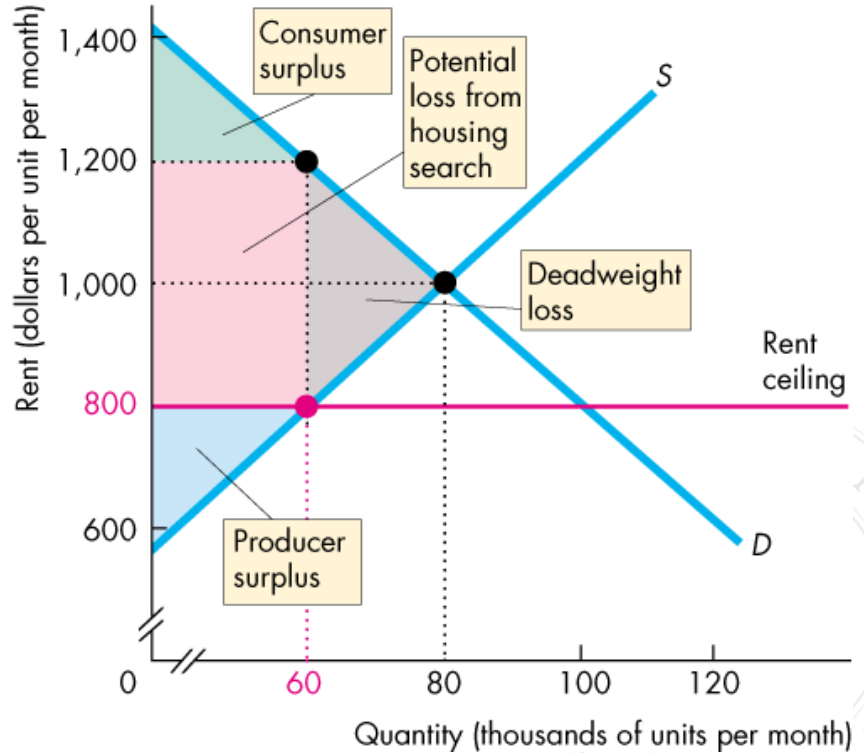
- A **black market** is an illegal market that operates alongside a legal market in which a price ceiling or other restriction has been imposed.
- A shortage of housing creates a black market in housing.
- Illegal arrangements are made between renters and landlords at rents above the rent ceiling—and generally above what the rent would have been in an unregulated market.



A Housing Market with a rent ceiling

Inefficiency of a Rent Ceiling

- A rent ceiling set below the equilibrium rent leads to an inefficient underproduction of housing services.
- The marginal social benefit from housing services exceeds its marginal social cost and a deadweight loss arises.
- A rent ceiling decreases the quantity of housing supplied to less than the efficient quantity.
- A deadweight loss arises.
- Producer surplus shrinks.
- Consumer surplus shrinks.
- There is a potential loss from increased search activity.



A Housing Market with a rent ceiling

Are Rent ceilings fair?

- From the point of view of *fair rules of exchange*, a rent ceiling is unfair because it blocks voluntary exchange.
- In terms of *results*, a rent ceiling is unfair because it does not generally benefit the poorest potential renters, i.e., the ones with the lowest willingness to pay because of their income.
- A rent ceiling decreases the quantity of housing and the scarce housing is allocated by
 - Lottery (random assignment)
 - First-come, first-served
 - Discrimination
- None of these methods leads to a fair outcome!



Price floors

A **price floor** is a regulation that makes it illegal to trade at a price lower than a specified level.

When a price floor is applied to labor markets, it is called a **minimum wage**.

If the minimum wage is set *below* the equilibrium wage rate, it has no effect. The market works as if there were no minimum wage.

If the minimum wage is set *above* the equilibrium wage rate, it has powerful effects.



A Labor Market with a Minimum Wage

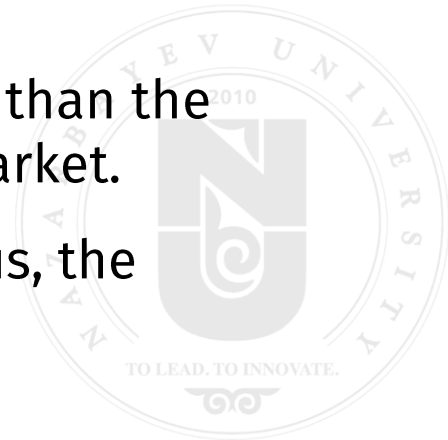
Minimum Wage Brings Unemployment

If the minimum wage is set *above* the equilibrium wage rate, the quantity of labor supplied by workers *exceeds* the quantity demanded by employers.

There is a surplus of labor.

The quantity of labor hired at the minimum wage is *less* than the quantity that would be hired in an unregulated labor market.

Because the legal wage rate cannot eliminate the surplus, the minimum wage creates unemployment.



A Labor Market with a Minimum Wage

The equilibrium wage rate is \$6 an hour.

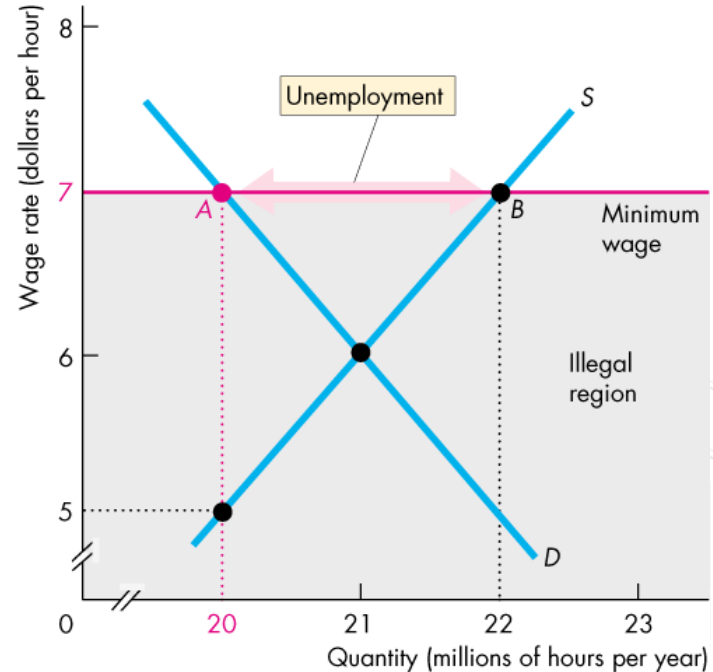
The minimum wage rate is set at \$7 an hour.

So the equilibrium wage rate is in the illegal region.

The quantity of labor employed is the quantity demanded.

The quantity of labor supplied exceeds the quantity demanded and unemployment is created.

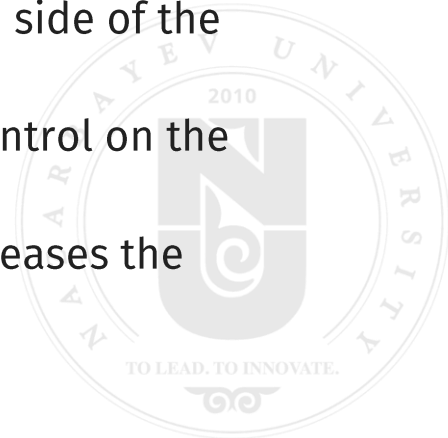
With only 20 million hours demanded, some workers are willing to supply the last hour demanded for \$5.



A Labor Market with a Minimum Wage

Is a minimum wage fair?

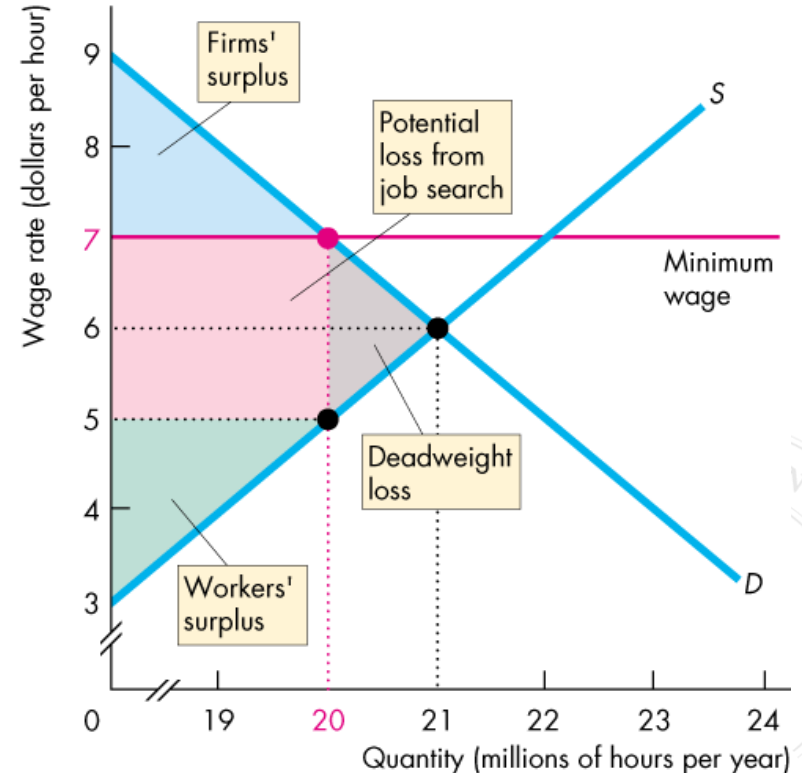
- This is a contested question with not an easy answer.
- On the one hand, it is a fact that the minimum wage restricts employment by introducing an artificially high wage level.
- Whether it creates unemployment, depends on the structure of the market in which the minimum wage is applied: the bargaining power of each side of the market pre-minimum wage is crucial.
- However, it is generally agreed that *ceteris paribus*, an artificial control on the wage will create unemployment.
- Specifically, most economists believe that the minimum wage increases the unemployment rate of low-skilled younger workers.



A Labor Market with a Minimum Wage

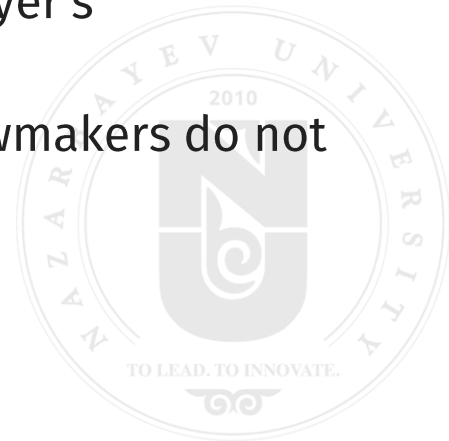
Inefficiency of a Minimum Wage

- A minimum wage leads to an inefficient outcome.
- The quantity of labor employed is less than the efficient quantity.
- The supply of labor measures the marginal social cost of labor to workers (leisure forgone).
- The demand for labor measures the marginal social benefit from labor (value of goods produced).
- A minimum wage set *above* the equilibrium wage decreases the quantity of labor employed.
- A deadweight loss arises.
- The potential loss from increased job search decreases both workers' surplus and firms' surplus.



Taxes

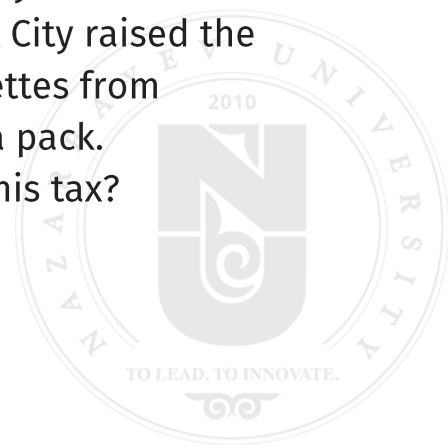
- Everything you earn and most things you buy are taxed.
- Who really pays these taxes?
- Income tax and the Social Security tax are deducted from your pay, and sales tax is added to the price of the things you buy, so isn't it obvious that you pay these taxes?
- Isn't it equally obvious that your employer pays the employer's contribution to the Social Security tax?
- It turns out it is not obvious who **actually pays** and that lawmakers do not decide who ends up paying.
- **Economic vs Legal Incidence.**
- Two types of taxes: per-unit and ad-valorem taxes.



Tax Incidence

Tax Incidence

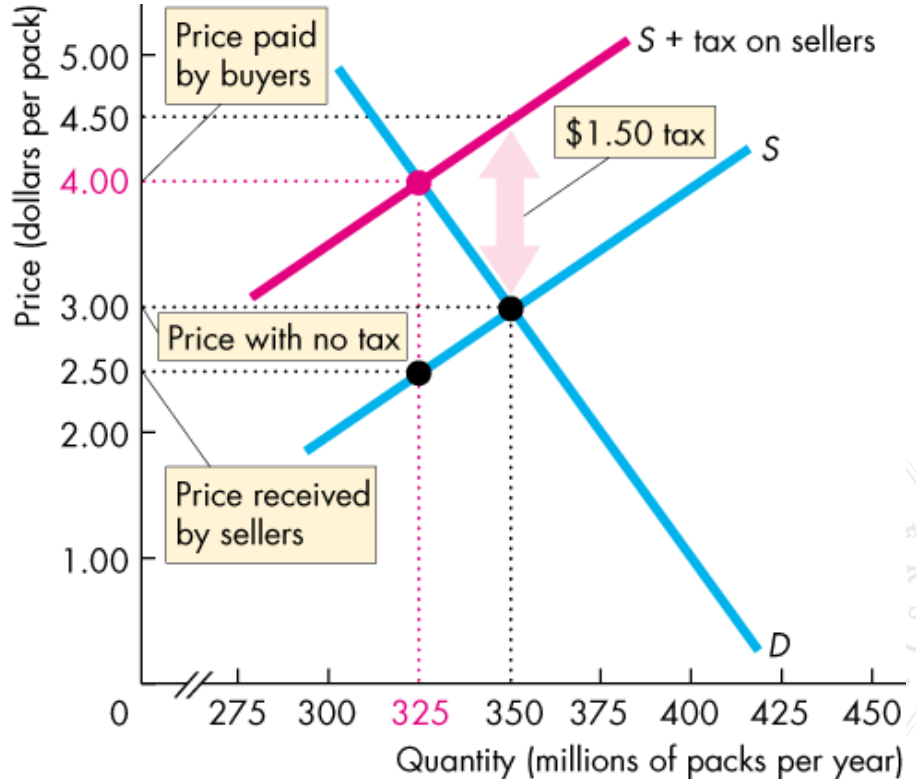
- Tax incidence is the division of the burden of a tax between buyers and sellers.
- When an item is taxed, its price might rise by the full amount of the tax, by a lesser amount, or not at all.
- If the price rises by the full amount of the tax, buyers pay the tax.
- If the price rises by a lesser amount than the tax, buyers and sellers share the burden of the tax.
- If the price doesn't rise at all, sellers pay the tax.
- Tax incidence doesn't depend on tax law!
- The law might impose a tax on buyers or sellers, but the outcome will be the same.
- To see why, we look at the tax on cigarettes in New York City.
- On July 1, 2002, New York City raised the tax on the sales of cigarettes from almost nothing to \$1.50 a pack.
- What are the effects of this tax?



Taxes

A Tax on Sellers

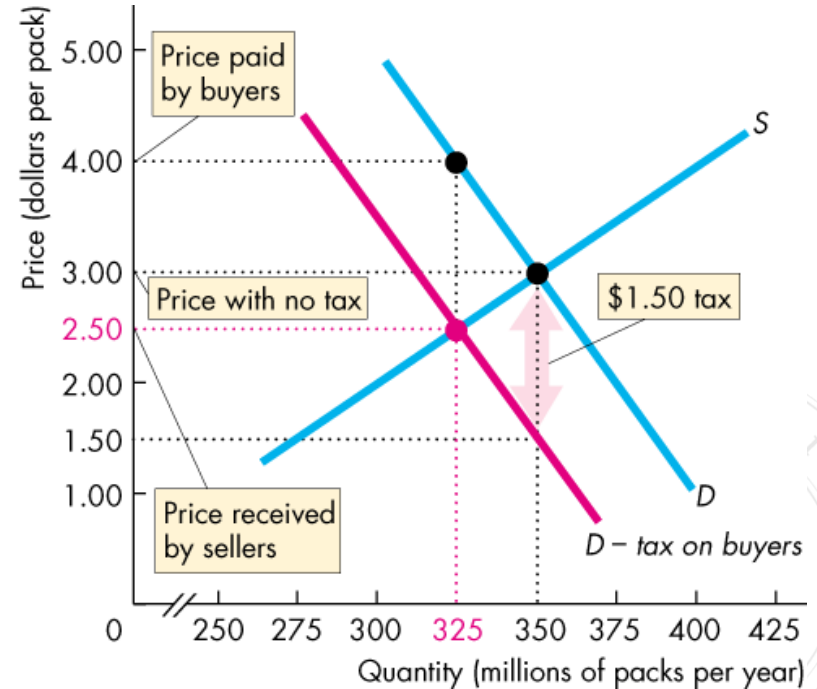
- The Figure shows the effects of this tax.
- With no tax, the equilibrium price is \$3.00 a pack.
- A tax on sellers of \$1.50 a pack is introduced.
- Supply decreases and the curve $S + \text{tax on sellers}$ shows the new supply curve.
- The market price paid by buyers rises to \$4.00 a pack and the quantity bought decreases.
- The price received by the sellers falls to \$2.50 a pack.
- So with the tax of \$1.50 a pack, buyers pay \$1.00 a pack more and sellers receive 50¢ a pack less.



Taxes

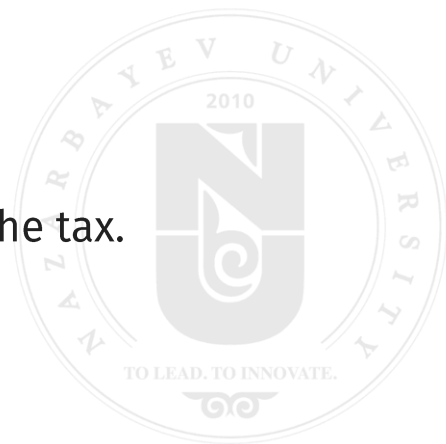
A Tax on Buyers

- Again, with no tax, the equilibrium price is \$3.00 a pack.
- A tax on buyers of \$1.50 a pack is introduced.
- Demand decreases and the curve $D - \text{tax on buyers}$ shows the new demand curve.
- The price received by sellers falls to \$2.50 a pack and the quantity decreases.
- The price paid by buyers rises to \$4.00 a pack.
- So with the tax of \$1.50 a pack, buyers pay \$1.00 a pack more and sellers receive 50¢ a pack less.



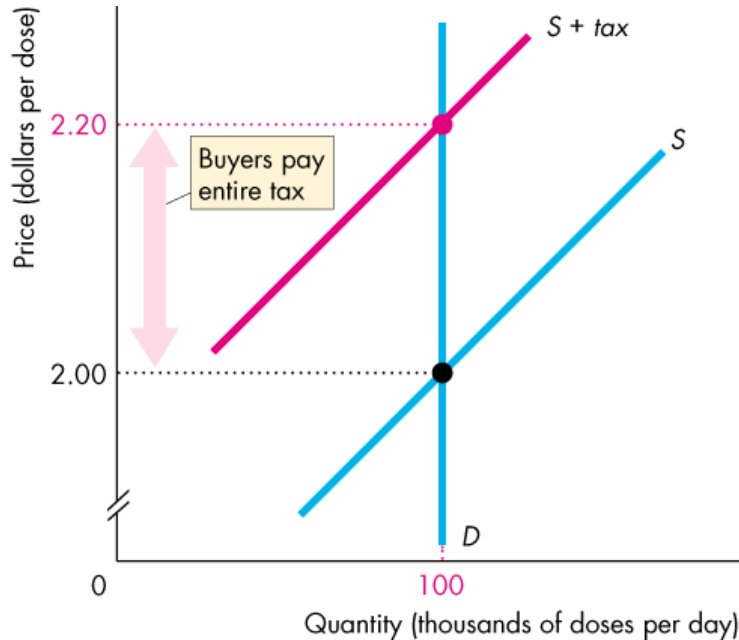
Taxes

- The lesson to learn from the previous example is that the **Economic tax incidence remains the same, irrespective of whether the legal incidence (law) says sellers pay or buyers pay.**
- The division of the tax between buyers and sellers depends on the elasticities of demand and supply.
- To see how, we look at two extreme cases.
 - Perfectly inelastic demand: Buyers pay the entire tax.
 - Perfectly elastic demand: Sellers pay the entire tax.
- The more inelastic the demand, the larger is the buyers' share of the tax.

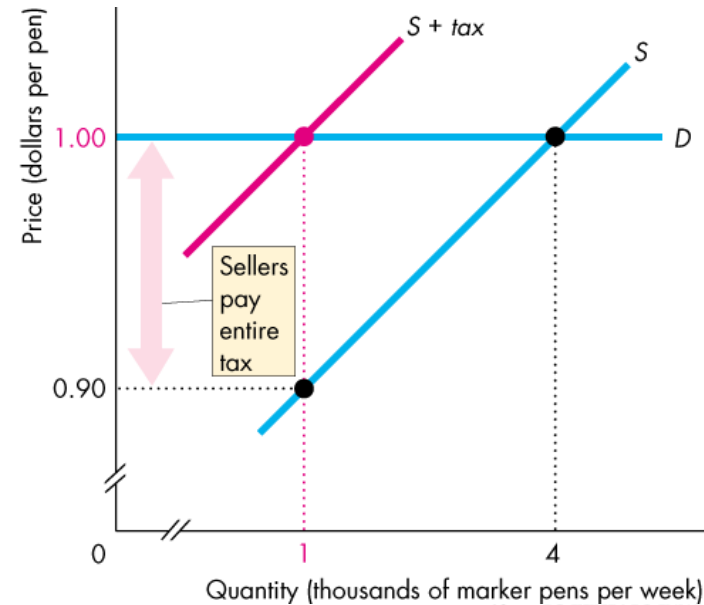


Taxes

Perfectly Inelastic Demand



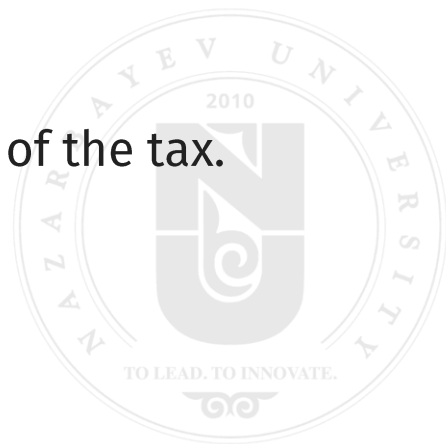
Perfectly Elastic Demand



Taxes

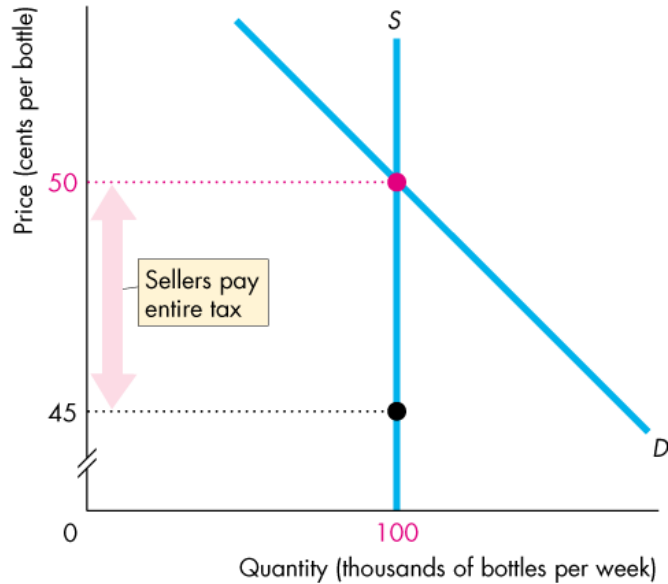
Tax Incidence and Elasticity of Supply

- To see the effect of the elasticity of supply on the division of the tax payment, we again look at two extreme cases.
 - Perfectly inelastic supply: Sellers pay the entire tax.
 - Perfectly elastic supply: Buyers pay the entire tax.
- The more elastic the supply, the larger is the buyers' share of the tax.



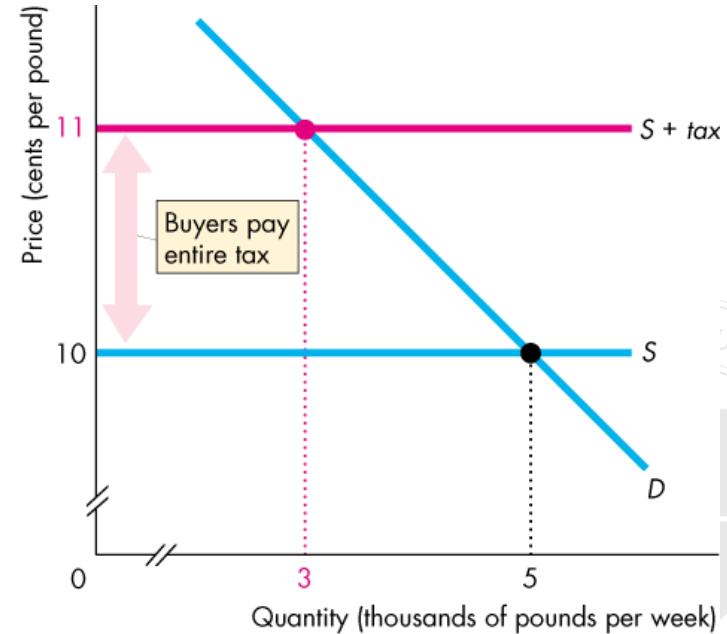
Taxes

Perfectly Inelastic Supply



(a) Perfectly inelastic supply

Perfectly Elastic Supply



(b) Perfectly elastic supply

Taxes

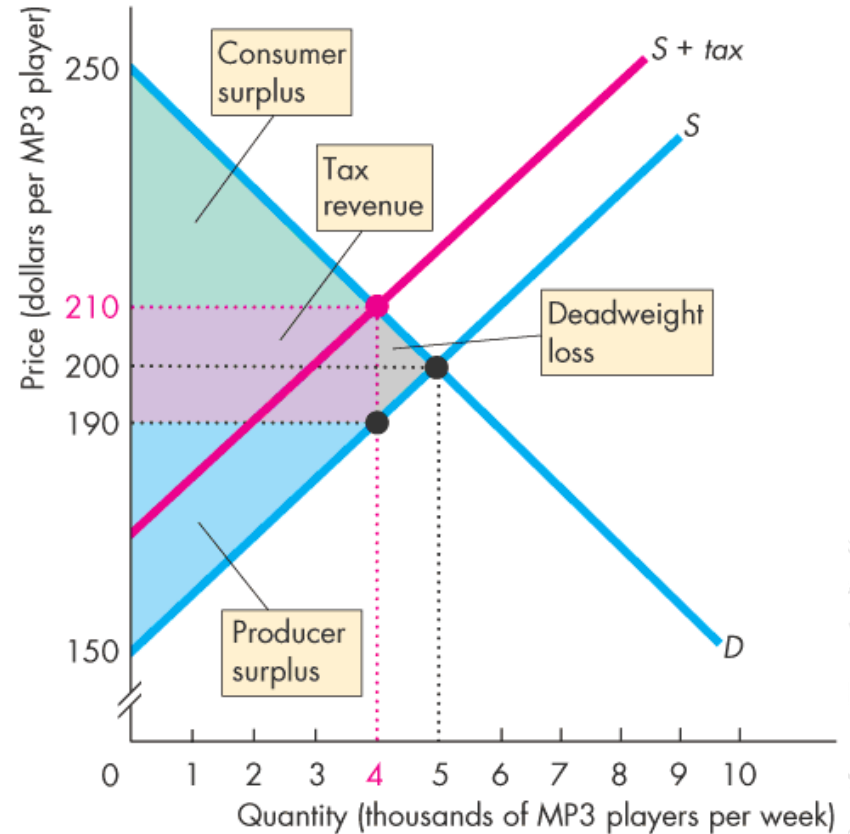
Taxes and Efficiency

- Whenever a tax restricts the quantity that is traded with respect to the unrestricted equilibrium price, an inefficiency will arise.
- Except in the extreme cases of perfectly inelastic demand or perfectly inelastic supply when the quantity remains the same, imposing a tax creates inefficiency.



Taxes

- With no tax, marginal social benefit equals marginal social cost.
- Total surplus (the sum of consumer surplus and producer surplus) is maximized.
- The market is efficient.
- The tax decreases the quantity, raises the buyers' price, and lowers the sellers' price.
- Marginal social benefit exceeds marginal social cost and the tax is inefficient.
- The tax revenue takes part of the total surplus.
- The decreased quantity creates a deadweight loss.



Subsidies and Production Quotas

- A **production quota** is an upper limit to the quantity of a good that may be produced during a specified period.
- A **subsidy** is a payment made by the government to a producer.



Subsidies

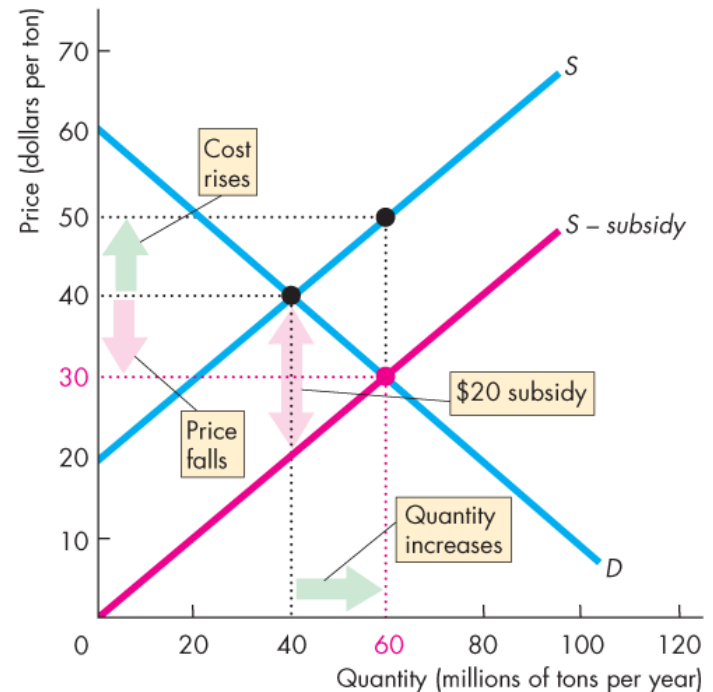
A subsidy of farm products

- With no subsidy, the price is \$40 a ton and 40 million tons a year are produced.
- With a subsidy of \$20 a ton, marginal cost minus subsidy falls by \$20 a ton and the new supply curve is $S - \text{subsidy}$.
- The market price falls to \$30 a ton and farmers increase the quantity to 60 million tons a year.
- But farmers' marginal cost increases to \$50 a ton.
- With the subsidy, farmers receive more on each ton sold—the price of \$30 a ton plus the subsidy of \$20 a ton, which is \$50 a ton.

Inefficient Overproduction

At the quantity produced:

- Marginal social benefit equals the market price, which has fallen.
- Marginal social cost has increased and exceeds marginal social benefit.



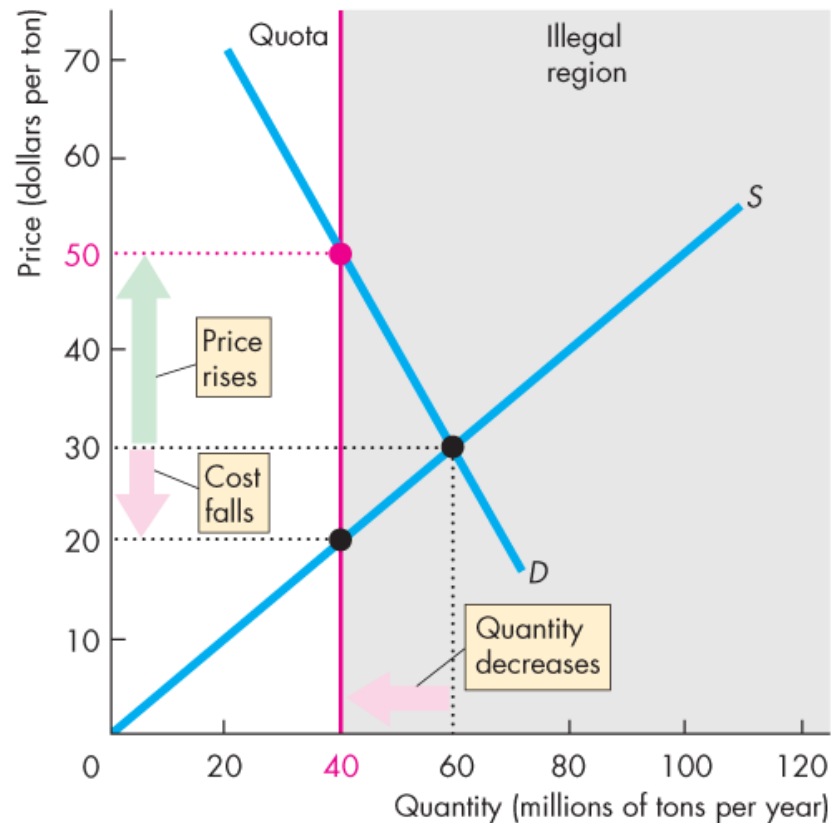
Production quotas

- With no quota, the price is \$30 a ton and 60 million tons a year are produced.
- With the production quota of 40 million tons a year, quantity decreases to 40 million tons a year.
- The market price rises to \$50 a ton and marginal cost falls to \$20 a ton.

Inefficiency

At the quantity produced:

- Marginal social benefit equals the market price, which has increased.
- Marginal social cost has decreased.
- Production is inefficient and producers have an incentive to cheat.



Markets for Illegal goods

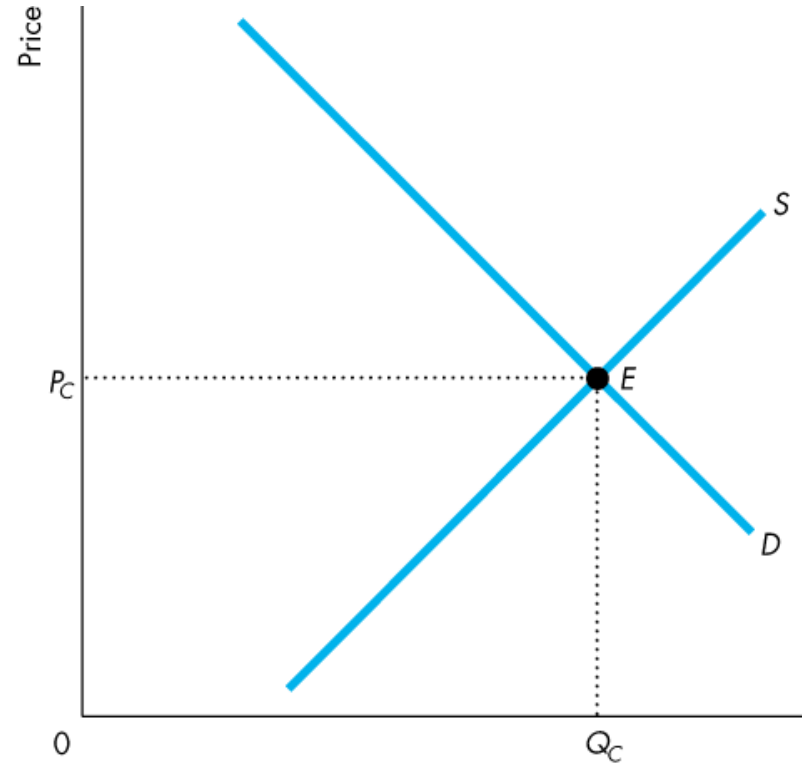
- Many governments prohibit the trade of some goods, such as illegal drugs.
- Yet, markets exist for illegal goods and services.
- How does the market for an illegal good work?
- To see how the market for an illegal good works, we begin by looking at a free market and see the changes that occur when the good is made illegal.



Markets for Illegal Goods

A Free Market for a Drug

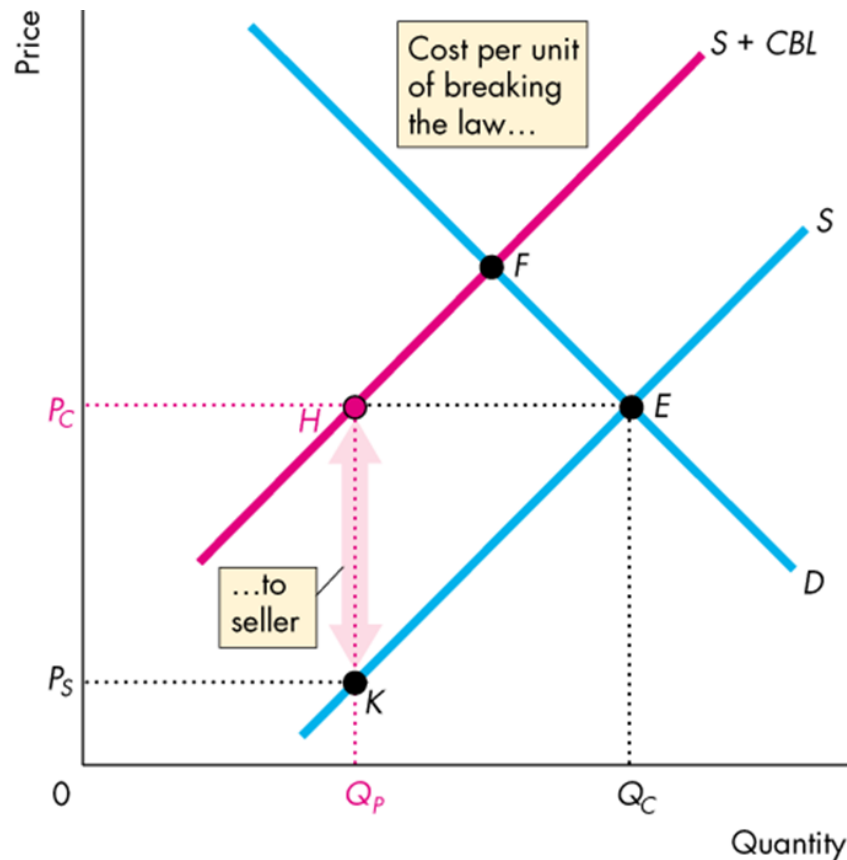
- The Figure shows the market for a drug such as marijuana.
- Market equilibrium is at point E.
- The price is P_C and the quantity is Q_C .



Markets for Illegal Goods

Penalties on Sellers

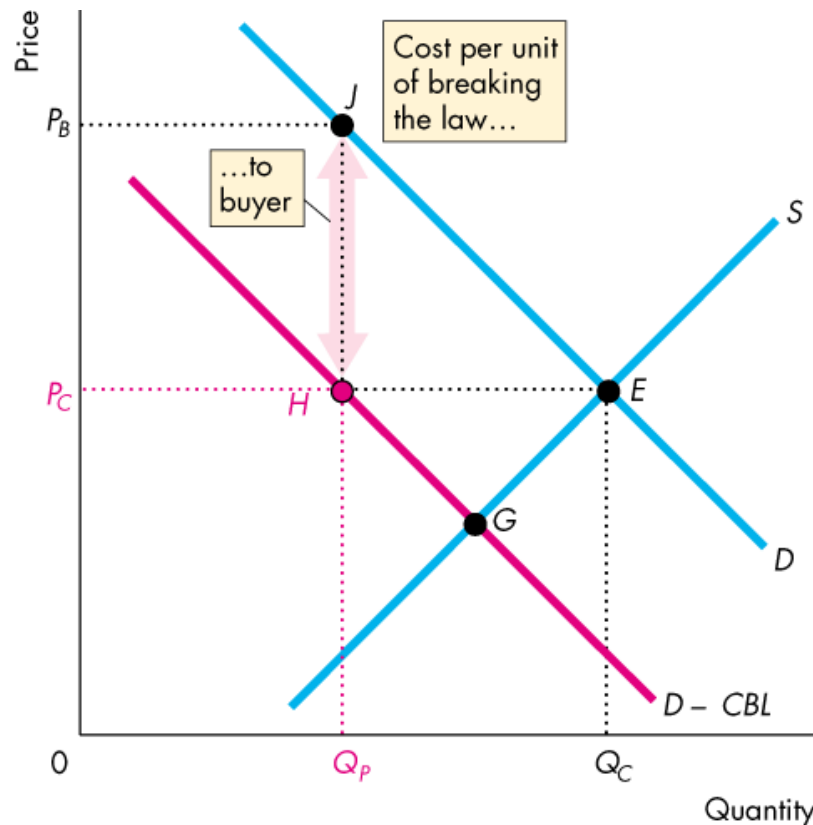
- If the penalty on the seller is the amount HK , then the quantity supplied at a market price of P_C is Q_P .
- Supply of the drug decreases to $S + CBL$.
- The new equilibrium is at point F . The price rises and the quantity decreases.



Markets for Illegal Goods

Penalties on Buyers

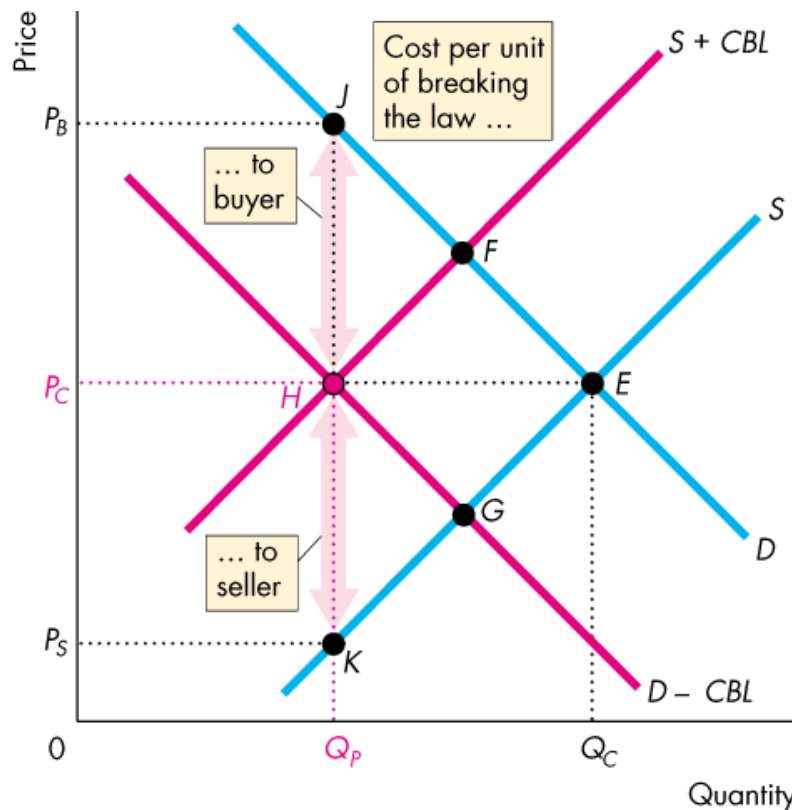
- If the penalty on the buyer is the amount JH, the quantity demanded at a market price of P_C is Q_P .
- Demand for the drug decreases to $D - CBL$.
- The new equilibrium is at point G. The market price falls and the quantity decreases.
- But the opportunity cost of buying this illegal good rises above P_C because
- the buyer pays the market price **plus** the cost of breaking the law.



Markets for Illegal Goods

Penalties on Both Sellers and Buyers

- With both sellers and buyers penalized for trading in the illegal drug, ...
- both the demand for the drug and the supply of the drug decrease.
- The new equilibrium is at point H.
- The quantity decreases to Q_P .
- The market price is P_C .
- The buyer pays P_B and the seller receives P_S .



Markets for Illegal Goods

Legalizing and Taxing Drugs

- An illegal good can be legalized and taxed.
- A high enough tax rate would decrease consumption to the level that occurs when trade is illegal.
- Arguments that extend beyond economics surround this choice.





End of Part I of this Lecture. Go to part II.
Chapter 6 of Parkin contains some of the material in this lecture.